Project Requirement Document

AI Recruitment Automation — Final Capstone Project (Option 2)

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Week / Day	Week 6, Day 31 ,Requirement Analysis
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1. Problem Statement

Manually screening job applications is slow and inconsistent. Recruiters spend significant time reading resumes, extracting relevant skills, and judging candidate fit against a role's requirements — a process that is prone to delay, reviewer fatigue, and inconsistent scoring across candidates.

2. Current Process & Pain Points

Current process: Candidates apply via email or ad-hoc forms. A recruiter manually reads each application, extracts skills and experience, and decides whether to shortlist, hold, or reject the candidate before contacting them individually.

Pain points:

- Applications arrive through multiple channels with no central, structured record
- Skills and experience are extracted manually from resumes/text — time-consuming and error-prone
- No consistent scoring rubric applied across candidates, leading to subjective decisions
- Recruiter has to manually track each candidate's status across the pipeline
- Candidate communication (acknowledgement, rejection, shortlist notice) is sent manually and inconsistently
- No structured audit trail of who was evaluated, when, and on what basis

3. Proposed Solution Outline

Trigger & Input: An n8n Form Trigger collects candidate applications (name, email, role applied for, experience, skills, resume/cover text).

Storage: A Supabase Postgres table stores raw submissions and processed candidate records.

Workflow steps:

- Form submission triggers a Code node that extracts structured candidate data
- An If node validates that required fields are present before proceeding
- A Gemini LLM chain extracts and scores candidate skills against the role's requirements
- A Code node parses and validates the LLM's structured output

- A rule-based assessment applies scoring thresholds to the parsed result
- The recruiter receives an email with a candidate summary for review
- A Wait node pauses the workflow pending human approval
- A Switch node routes the candidate to Shortlist, Reject, or Hold based on the recruiter's decision
- The candidate is notified automatically by email based on the outcome
- A final Postgres update records the candidate's status and decision for audit purposes

AI role: Gemini is used for skills extraction and fit scoring; the recruiter retains the final decision (human-in-the-loop) for every shortlist, hold, or reject action.

Business rules: A minimum skill-match threshold must be met before a candidate reaches recruiter review; human approval is mandatory before any candidate is finally shortlisted or rejected.

Data storage: Supabase Postgres holds candidate records, scores, statuses, and timestamps for a complete audit trail.

Notifications: Gmail SMTP sends recruiter review alerts and candidate status emails.

Error handling & security: Structured validation at each stage, retry logic for LLM/API calls, and logging of failures for review. Credentials are stored securely in n8n's credential store; no sensitive data is exposed in workflow logs.

4. Success Criteria

- The end-to-end workflow runs without manual intervention except the required human approval step
- Candidate data is reliably stored and queryable in Supabase
- Skills extraction and scoring are consistent across test candidates
- The recruiter receives clear, actionable summaries to support decision-making
- Candidates receive timely and accurate status emails
- A full audit trail exists for each candidate's journey through the pipeline
- The workflow handles normal, edge, and failure cases (e.g., missing fields, malformed LLM output, failed email delivery) without breaking